Development and Application of a Fully Autonomous Robot for Disaster Management and Humanitarian Purposes

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Abstract – This research paper presents the design and development of a fully autonomous robot used for disaster management and humanitarian purposes. The paper shows the necessity for such a robotic system, the challenges it addresses, and the technology used in its making. Disasters can occur at any moment of time in any corner of the world causing fearful situations including casualties of the victims as well as the people who go to rescue them. Post-disaster control has become one of the most tedious jobs to perform. Implementation of rescue missions is challenging due to safety and economic issues. Another big problem is the lack of time as the conditions on the site are very critical and unpredictable. The rescuers enter such zones to help the victims but in return face severe injuries due to the totally unknown environment which was created after the disaster. The robot does the task for the rescuers by entering the area and delivering food and water packs for the victims till the rescuers plan a practical mission for saving the victims. This requires no human intervention as the robot is fully autonomous. The real-world applications of such robots for disaster management and their effectiveness in improving the safety and reliability of post-disaster rescue technology have been discussed in this study.

INTRODUCTION

Disasters surely cannot be controlled by humans but how we act after the disaster is what makes a difference. After any disaster like earthquakes, landslides, collapsed buildings, tunnels or mines there may be people stuck inside rubble and debris formed due to the disaster. These sites are chaotic, dangerous and time-critical and human rescuers who enter these sites often face significant risks, including falling rubble, hidden voids, dangerous air quality, and the constant threat of aftershocks or secondary collapses [1]. The first few hours after the disaster are very critical – victims trapped under or behind the debris may be seriously injured, dehydrated, and even hungry and even the slightest delay could be the difference between life and death [2]. The rescuers must move slowly and cautiously which delays the supply of essential aid [3]. In some cases, the rescuers can't even access the affected zone due to small and hazardous pathways [4]. Until the rescuers plan and implement the rescue mission there is a need to supply essential aid to the victims quickly and safely. (Figure 1 a [5] and b [6] show the conditions of some rubble filled zones after a disaster)



(a) Collapsed mine



(b) Collapsed tunnel

FIGURE 1. Post-disaster conditions – rubble filled sites

This paper proposes a small, four-wheeled, fully autonomous disaster management robot which can move easily across rubble–filled terrain. It possesses autonomous navigation capabilities without any human intervention. It is equipped with a thermal camera, which helps it navigate to the victims through the heat dissipated from them. Additionally, it is equipped with a high–tech camera system which provides a real–time video feed and enables advanced image processing for performing tasks such as identifying objects and assessing the environment. It can move through rubble filled areas because of its larger wheelbase and wheel size. The robot also possesses a dropping mechanism, which is a conveyor belt with divisions at uniform distances to control the drop rate of the food and water packs. It has the capacity to store four small packs and two big packs (if required) during the rescue mission on the conveyor belt itself. This helps it to save time while helping the victims. The packs must be placed prior to the rescue mission so that the robot can complete the mission in one run. If required, it can come back at the control point to collect more packs.

The primary objective of our robot is to navigate through rubble–filled zones without significant difficulties, detecting and classifying victims and objects using heat and colour sensing, and delivering food and water packs at those locations with the help of a reliable dropping mechanism.

There are very few robots working for this purpose currently. Figure 2 (a [7], b [8], c [9], d [10]) shows the different types of rovers used for disaster management, currently across the world. These robots have slightly different applications as they have been used for different types of post-disaster rescue missions.



(a) Miniature Robot



(b) Quince Robot





(c) Reconfigurable Wheeled Robot **(d)** Water Canon Robot

FIGURE 2. Examples of some real-world disaster management robots

METHODOLOGY

1. Mechanics of the Robot:

The mechanical design of the robot has been developed keeping in mind the conditions of the terrain in which the robot needs to move throughout the rescue mission. The robot consists of a strong body, drive system, and conveyor belt for payload delivery, and protection enclosures.

1.1 Chassis Design: The chassis is made of mild steel (2 mm to 3 mm thickness). Mild steel has been chosen as the base material for the whole structure of the robot because it provides high durability and strength, especially in uneven terrain. The structure has been designed in a 2-level arrangement:

- **Lower-level –** It contains the motors, wheels, motor drivers, battery and electronics. Enough space of 10 cm has been given between the first and second levels for proper organization of the electronic components.
- **Upper-level –** It contains the dropping mechanism, which is a conveyor belt that has divisions on it which control the drop rate of the items.

1.1.1 Protection Plate:
A plate with dimensions matching the chassis (lower-level and upper-level plates) has been placed below the motors to protect them from collisions with rubble. This provides quite a good ground clearance of about 4 – 5 cm, which is enough for a rubble–filled terrain. The robot can easily overcome small rubble without any collisions. The motors are placed between the lower-level plate and the protection/bash plate so that the heat from the motors can easily be managed. Motor drivers can also be placed in the same space if required.

1.1.2 Support for upper plate:
The upper–level plate is supported by four pillars or square pipes (hollow) made of mild steel and of the same thickness as that of the plate. These pillars are fitted with screw from both ends. They are placed on the four corners of the lower–level plate for better weight distribution.

1.2 Locomotion System: The robot is driven by a 4-Wheel Drive (4WD) system which has large 130 mm x 60 mm wheels. These wheels have considerably deep grooves, which help in improving the grip. This provides good traction and torque as well. Larger wheels also contribute to the improved ground clearance, which ensures that there is no collision with rubble under the vehicle. Also, the wheels have a very low chance of getting stuck in rubble. The wheelbase is also larger, which adds to stability.

1.2.1 Wheels as indirect suspensions:
The wheels are filled with foam, which absorbs the shocks while moving and acts as a sort of absorber or suspension. If required, springs can be fitted with the motor mounts to provide good suspension to the vehicle.

1.2.2 An alternative to wheels – 'Tracks'

Tracks can also be used in place of wheels, as tracks do not require a separate suspension system. They have a timing belt wrapped around two or more wheels or rollers, which can wrap around rubble and easily climb or overcome the rubble without getting stuck. It can also save the robot from toppling.

1.2.3 Adjusting the Centre of Mass:

The Centre of Mass (COM) of the robot will be as low as possible and probably in the Line of Gravity. This would further help in stabilizing the robot and saving it from toppling. This can be achieved by placing the battery between the lower–level plate and the protection plate, given its considerable weight. The wheels are 130 mm in diameter, and the total height of the robot is about 295 mm, which is almost double the size of the wheels (without payloads). This ensures that the robot will not topple or face any difficulty in overcoming obstacles and navigating throughout the area.

1.3 Motors and Transmission: The wheels are driven by DC geared motors with a high gear ratio to attain more torque. Heavy brackets have been used for mounting the motors which are attached to the wheels using couplers. The motors have been selected based on torque and speed calculations, which are as follows:

a) Torque Calculation:

$$T = \frac{W \times r \times \mu}{n}$$

Where, W = weight of the robot = 12 Kg x 9.81 m/s^2 = 118 N (Approx.)
r = radius of the robot = 0.065 m
μ = rolling resistance factor (0.25 for rubble/rough terrain)
n = number of wheels = 4 (all driven)
T = 0.48 Nm (Approx.)

Hence, each motor must provide at least 0.5 Nm (5 Kg-cm) torque.

Now, with a safety factor = 2, motors of greater than or equal to 1.0 Nm (10 Kg-cm) torque may be selected.

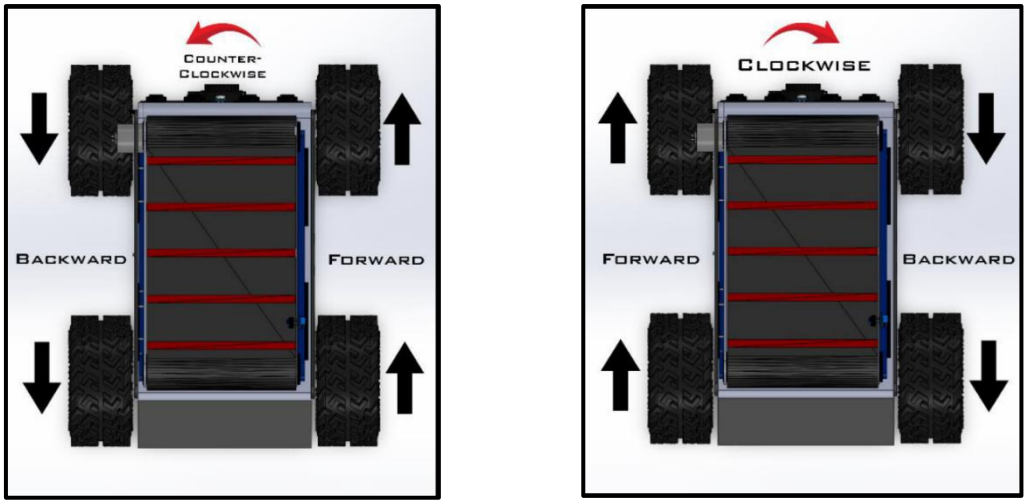
b) Speed Calculation:

$$v = \frac{2 \times \pi \times r \times N}{60}$$

Where, r = wheel radius (m)
N = motor rpm under load
For r = 0.065 m and N = 60 rpm, v = 0.41 m/s (1.5 Km/h) (Approx.)
This speed is very suitable for navigating through a rubble-filled zone.

1.3.1 Mechanism for taking turns:

The motors are paired depending on the angle by which the robot needs to make turns, such as 90°, 180°, or 360° turns. For instance, when two motors on the right side of the robot turn towards the back, and the other two turn in the opposite direction, the robot takes a right turn by 180°. These maneuvers help the robot to detect and classify the victims using the camera and sensors placed in front of the robot. If tracks are used, then there will be side plates to avoid rubble from entering the inner parts of the tracks. The turning mechanism of the robot has been shown in figure 3 (a, b).



(a) Turning left **(b)** Turning right

FIGURE 3. Turning mechanism of the robot

1.3.2 Use Case:

A 360° turn is made when the robot starts its mission, to scan the area and find the victims. Once the drop location has been determined, the robot, after reaching there, will make a 180° turn to drop the payload (food and water packs) near the victim, always (standardized procedure). The robot can also make a pivoted turn if required to avoid an obstacle while navigating. The motors also allow the robot to move in the reverse direction.

1.4 Dropping Mechanism: The robot uses a conveyor mechanism fitted in the second level, above the upper plate. The conveyor has a DC motor fitted to one of the rollers (powered) to move it, which causes the second roller (unpowered) to move in the same direction as they are connected via a belt, which has enough grip to hold the payload over it. Moreover, the conveyor features plates made of mild steel or aluminum on three sides, which act as walls to prevent the payload from falling due to the robot's movement. The payload will be dropped at the back of the robot after it makes a 180° turn, so that the fallen payload does not obstruct the path of the robot. There is a slight incline placed just at the edge of the upper plate for the payload to fall a bit far from the robot, so that there is no complication while leaving the spot.

The rollers may be 3D printed to have a specific design and shape. Ready-made rollers, which are compatible with a good timing belt, can also be used.

1.4.1 Controlling the rate of drop:

There are divisions on the belt which move along with it. The purpose of doing this is to control the rate of drop of items. The time it takes for one division to drop an item can be calculated, and a system can be set up so that every victim gets an equal number of items. An IR sensor can also be placed at the end from where the items fall to check whether the item has been dropped successfully. This IR sensor can give signals to Arduino Mega, which in turn can control the working of the DC motor fitted to the conveyor.

1.4.2 Divisions on the conveyor belt:

The conveyor has been mounted on the upper plate such that the divisions are not obstructed while they move continuously along with the belt. The conveyor belt has a total of six divisions, which are separated by wedges. These divisions have been grouped according to the sizes of the items which need to be carried:

- **4 Divisions (Group 1) –** These divisions have a smaller area and hence can carry smaller items, but four items at a time, so a greater number of victims can be targeted.
- **2 Divisions (Group 2) –** These divisions have a larger area and hence can carry larger items, but only two items at a time, so relatively a smaller number of victims can be targeted. The

group can be decided based on the size of the items which need to be dropped. The conveyor can carry about 1 – 3 Kg of payload at a time.

1.5 Ground Stability Analysis: The stability of the robot can be verified by calculating the ground pressure, which can be given as follows:

$$P = \frac{W}{A}$$

Where, W = 118 N

A = total contact area = contact area of wheels = 4 x (0.13 x 0.06)

$$= 0.0312 \text{ m}^2$$

$P = 3782 \text{ N/m}^2$ (Approx.)

This pressure is not that high. This ensures that the wheels do not get stuck in the rubble, providing better stability.

1.6 Protection and Safety: All the electronic components are enclosed inside a shielded compartment. If a collision takes place, the first component to come in contact with the obstacle will be the wheel, and hence it will not affect the strength of the robot. Bumpers have been added for protection. Dampers have been used for protecting the electronic components from vibrations. The kill switch has been kept accessible so that it can be pressed instantly in case of emergency.

Since the plates are made of mild steel, they are appropriately wrapped with an insulation material for the prevention of short circuits, because mild steel can conduct electricity. All the wires are properly insulated, and the connections are made firm by soldering so that they do not break in the middle of the rescue mission. Holes have been made in the structure for managing the heat emitted from components. Washers and nuts have been used to reduce wear-off of the metal. The battery has been colored brightly so that it can be seen or extracted easily during an emergency.

2. Electronics of the Robot:

The electronic design of the robot has been made such that it can perform the required tasks with maximum accuracy and precision. It should also be energy and time efficient. Special considerations have been made regarding the autonomous nature of the robot, and accordingly, connections have been made. The robot consists of two processing boards, sensors, a camera, etc.

2.1 Processing Units (Autonomy): The robot is equipped with 2 very good processing units – Arduino Mega and Raspberry Pi 5 – which perform specific tasks required to make the robot autonomous and valuable for the application selected.

2.1.1 Arduino Mega:

Arduino Mega [11] is a microcontroller board which performs various tasks in the robot. It performs all the low-level tasks like controlling the motors, encoders, ultrasonic sensors, and the kill switch (failsafe mechanism). It communicates with the Raspberry Pi 5 board using various protocols, such as UART (Universal Asynchronous Receiver/Transmitter), I2C (Inter–Integrated Circuit), and SPI (Serial Peripheral Interface). They can also be interconnected by simply using a USB cable.

The Arduino receives data from the R Pi and sends signals to the components under its control. It works on 5V, so there is a requirement for a DC – DC Buck Converter, which makes sure that a stable 5V DC supply is given to the board.

It is responsible for all the maneuvers performed by the robot, such as 180° or 360° turns, etc. It is also responsible for detecting obstacles using the ultrasonic sensors placed in front of the robot. If the robot encounters an obstacle that cannot be overcome, the motors are disarmed, and an alternate path is taken.

2.1.2 Raspberry Pi 5 (8 GB RAM):

Raspberry Pi 5 [12] is a microprocessor board which performs various tasks in the robot. It looks after functions like vision and navigation using the Pi HQ camera, heat sensing using the MLX90640 thermal camera/sensor, using high–level algorithms. It also controls the ICM-42688-P.

It takes live feed from the Pi HQ camera, which helps the robot in detecting the victims using a set of colors in real time. Once the victim is detected, the thermal camera (MLX90640) senses the heat coming from various places in its FOV. It selects the spot which is detected as the hottest in terms of temperature. The reading of the thermal camera is used to double-check the findings of the Pi HQ camera, ensuring that the robot has a reliable location to move to for the dropping task.

2.2 Motors and Motor Drivers: The motor we have chosen (Cytron MD 30C 30A) [17] provides a very good torque of about 40 Kg–cm at 60 rpm. This much torque is enough for the rubble–filled surface. The motor drivers are compatible with the selected motors and can handle the amount of current required for the above-mentioned high torque. A fuse has been used for safety reasons. A 20 KHz PWM signal will be used for communication.

2.2.1 Motor Drivers and Encoders:

Motor drivers are connected to Arduino Mega, which is the microcontroller. Arduino controls when the motors will start and when they will stop. Motor drivers act as channels through which the motors receive voltage and current (in short power) to run at certain speeds (rpm). The selected motors do not have encoders inside them, so we have used external encoders, which work to bring stability in the robot's movement. They are also associated with PID and odometry.

For the motors selected:

- **Continuous current** = 30 A (Approx.)
- **Peak current** = 80 A (Approx.)

Accordingly, motor drivers have been chosen based on their current ratings.

2.3 Sensing: The robot is equipped with Pi HQ camera [13], MLX90640 [14] thermal camera/sensor, ICM–42688 IMU [15] and A02YYUW [16] ultrasonic sensors. All these sensors or cameras make sure that the robot moves in the correct direction, that is, towards the victim to drop the payloads.

2.3.1 Obstacle detection and avoidance with the help of ultrasonic sensors:

The ultrasonic sensors in the robot help detect obstacles that cannot be overcome, thereby determining the distance between the robot and the obstacle. This allows the robot to reach the drop location without colliding. As soon as an obstacle comes within a specific range of the robot, the sensor immediately gives the information to Arduino Mega, which in turn stops the motors from running. This prevents the robot from colliding. The robot then finds an alternate path for reaching the desired drop location. These sensors are waterproof and work very well in rubble.

The basic working of an ultrasonic sensor has been shown in figure 4 below.

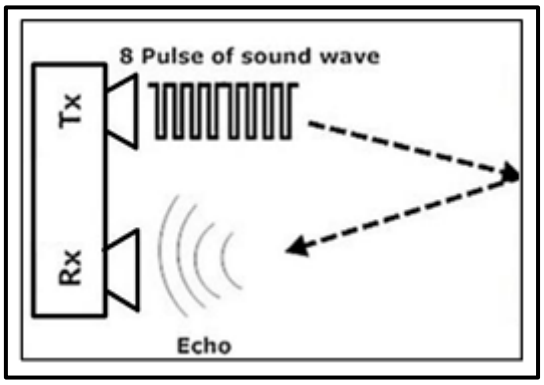


FIGURE 4. Working of an ultrasonic sensor

2.3.2 The Role of IMU:

The IMU sensor must be placed at the center of rotation of the robot so that when the robot makes a pivoted or 360° turn, the IMU ensures that there is no or almost negligible drift. This means that the rover's position is not changed, and hence the displacement is zero. IMU has inbuilt accelerometers and gyroscopes, which are used for balancing the robot. The selected IMU does not have a magnetometer to avoid interference with the motors. IMU and Encoders work collaboratively by taking care of the position and stability of the robot. This ensures that there is no displacement or wobbling. This also helps one when using PID for better and steadier movement.

The conveyor on the second level has an IR sensor at the end, which is useful for detecting whether an item has been dropped or not. The sensor gives a feedback signal to the Arduino, which disarms the conveyor motor after the item is dropped.

2.4 Object Detection and Classification: Pi HQ camera and MLX90640 thermal camera are the two components which are used for detecting the victims in the area.

2.4.1 Object Detection with the help of Pi HQ camera:

Once the rescue mission starts, the robot makes a 360° turn to explore and scan the surrounding environment. The Pi HQ camera, which is fitted in the front, is connected to the R Pi 5. It sends real–time feedback to the R Pi. During the scan, it detects the victim using a set of defined colors, as it is an RGB camera. This detection is considered by the thermal camera (MLX90640).

2.4.2 Cross–checking with the help of thermal camera:

The thermal camera scans the area under its FOV and finds the hottest spot in the region. This determination helps R Pi to cross–check the detection of the Pi HQ camera. If the two interpretations are nearly aligned, the robot begins moving in that direction with live obstacle avoidance. Once it reaches the drop point, it starts the procedure for dropping the payloads. Before every task, the buzzer will beep to indicate the completion of a particular action. Hence, the Pi HQ camera does object detection, and the thermal camera confirms the detection. A signal is given to the Arduino that the robot must start moving in the determined direction to complete the required tasks.

2.4.3 Algorithms used for object detection and classification:

R Pi uses various algorithms to perform object detection and classification using the components mentioned above.

- **Preprocessing:** It is easier to detect colors using HSV as compared to RGB. Noise in the mask is removed by morphological filtering.

- **Object Detection:** Contour detection does the extraction of the shapes of blobs from the mask. The small blobs are totally filtered out using the method of area thresholding.

- **Classification:** For color–coded victims, just color and shape filters are used. There is no

Machine Learning required, so there is no need to train a model. This is because it runs in real time.

- **Decision Layer:** If a victim is detected and classified correctly, then the distance is calculated using the location of the pixel. Then the coordinates of the target are passed to navigation logic, after which the robot drives to the point where the payload needs to be dropped.

2.5 Power Electronics: The robot needs two batteries to supply enough power to every component so that they function correctly. The batteries provide a power supply to all the components for approximately 30 minutes or even an hour, allowing the robot to cover the whole area in a single charge without compromising efficiency.

- The first battery will be solely used to power the four motors, as they require high torque, due to which the current draw increases. So, the battery is directly connected to the motor drivers.
- The second battery will give power to Arduino and R Pi, which further provide power to the conveyor motor, conveyor sensor (IR), Pi HQ camera, thermal camera, ultrasonic sensors, etc.

DC – DC Buck Converters are used to supply a constant 5 V DC voltage to Arduino and R Pi.
As a result, both the boards receive a stable voltage supply of 5 V independent of the 12 V from the battery.

This power distribution will help in increasing the efficiency and organization of the circuit in the robot. The robot also uses bus bars for proper organization of the wires.

Battery Runtime Calculation:

Total Power = Power for electronic components + Power for motors
Runtime = Energy/Power

2.6 Safety elements: All the electronic components must be kept safe from sudden voltage fluctuations from the battery, so buck converters are used for specific components which require constant voltage for working. A kill switch is placed on the robot, which is easily accessible. This must be pressed immediately in case of an emergency so that the motors are disarmed and the situation is under control. Moreover, fuses are used to manage the overload from the battery. Bus bars are used for the organization of the wires correctly. The positive and negative terminals must be kept away from each other, because if they meet, then there will be a short circuit. Wires with specific thickness must be used so that they can handle different amounts of current passing through them. Also, the wires must be properly insulated. The circuit must be tested and verified before performing the tasks to ensure there are no component issues during the mission. The detailed circuit diagram of the robot is presented in Appendix 1.

The complete design of the robot has been shown in figure 5 below:

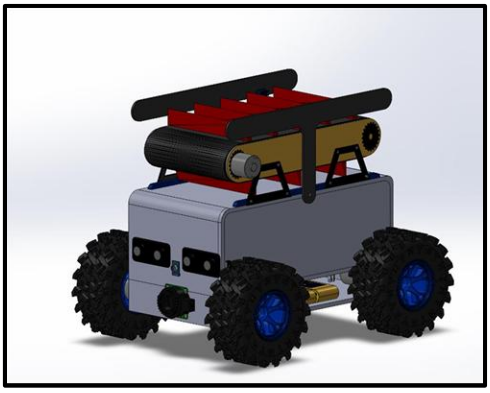
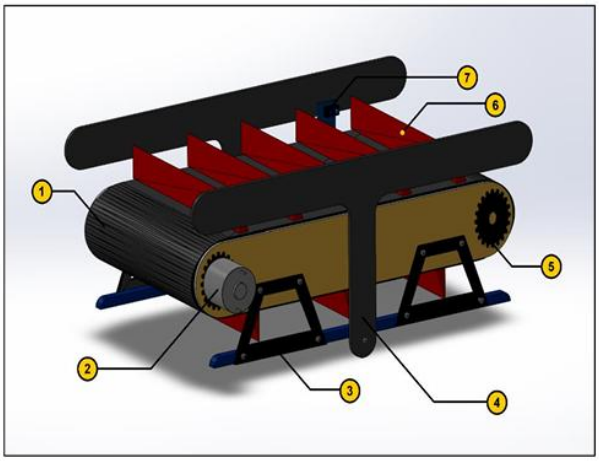


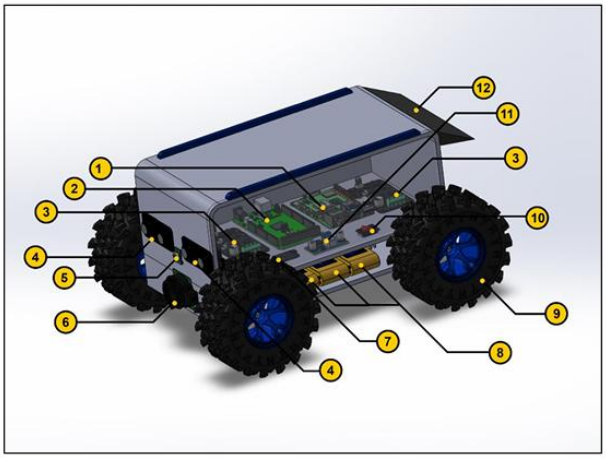
FIGURE 5. Isometric view of the robot

Labelling of different parts of the robot has been shown in figure 6 (a, b) below:



Sr. No.	Component
1.	Conveyor Belt
2.	DC motor for driving the belt
3.	Support for the Conveyor Belt (Dropping mechanism)
4.	Side plate for protection (Barrier)
5.	Rollers with grip (grooves)
6.	Divisions (Wedges)
7.	IR Sensor

(a) Labelling of the upper/second level



Sr. No.	Component
1.	Raspberry Pi 5 (8 GB RAM)
2.	Arduino Mega
3.	Motor Driver Module (Cytron MD30C 30 A)
4.	Ultrasonic Sensor (DYP–A02YYUW)
5.	Thermal Sensor (MLX90640)
6.	Pi HQ Camera
7.	IMU (ICM–42688–P)
8.	Batteries
9.	Wheel
10.	Switch
11.	DC – DC Buck Convertor
12.	Ramp (Part of dropping mechanism)

(b) Labelling of the lower/first level

FIGURE 6. Labelling of different parts of the robot

CONCLUSION

The world is full of things which are totally unpredictable and uncontrollable by humans. Disasters come on the top of this list as they can neither be predicted nor be stopped. They occur, they destruct and they vanish. Due to this uncertainty, there is a need for the world to show preparedness for the actions they need to take after a disaster hits, because this is in our control. The consequences of these actions depend on how efficiently we act during these times. Disasters like earthquakes, mine accidents, tunnels getting collapsed, etc. may cause injuries to the people present over there and it may lead to a need of immediate delivery of emergency aid to them. The rescuers may become victims of the aftershocks so a fully autonomous disaster management robot can be assigned to do the job because it does not require any human intervention while working. These robots are very compact and energy efficient. They deliver in no time and can be used continuously for hours without stopping. This indeed fulfils a very crucial humanitarian purpose of post-disaster rescue.

APPENDIX 1

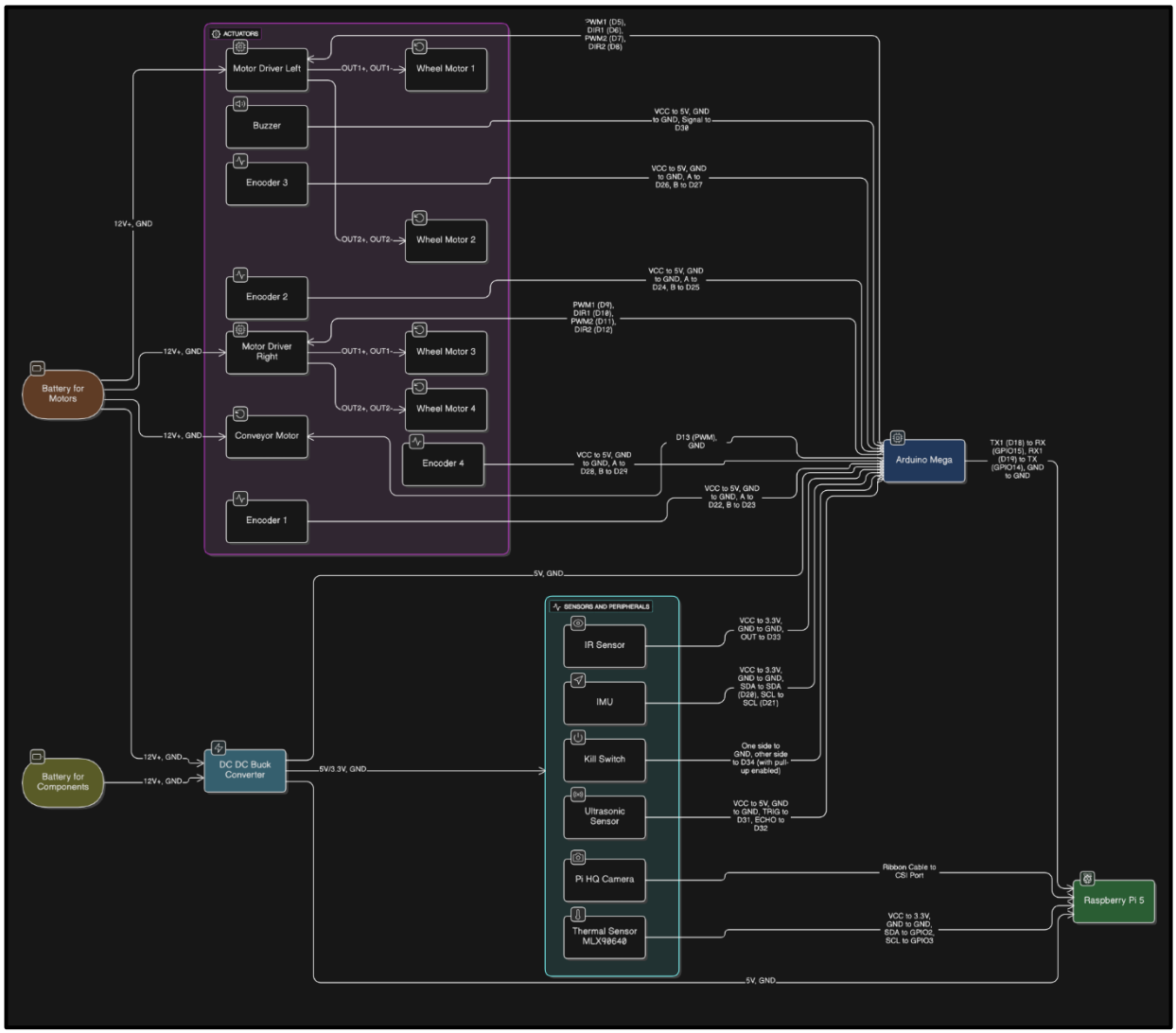


FIGURE 7. Complete circuit diagram of the robot

REFERENCES

1. H. M. Ghodsi, N. Sohrabizadeh, and A. Khani Jazani, "Challenges Faced by Medical Rescue Teams During Disaster Response: A Systematic Review Study," *Disaster Med Public Health Prep*, vol. 15, no. 3, pp. e364–e373, Jun. 2021.
2. J. S. P. D. L. V. Alagawadi, "The Golden Hour: A critical time between life and death," *Think Global Health*, Mar. 2023.
3. *Detecting, Locating, and Characterizing Voids in Disaster Rubble for Search and Rescue,* ResearchGate, 2019. [Online]. Available: https://www.researchgate.net/publication/335352713_Detecting_Locating_and_Characterizing_Voids_in_Disaster_Rubble_for_Search_and_Rescue
4. *DHS Shoring Guidebook: Structural Engineering for Disaster Management.* U.S. Department of Homeland Security, Sep. 2011. [Online]. Available: https://www.dhs.gov/xlibrary/assets/st/st-120108-final-shoring-guidebook.pdf
5. "Death toll rises to 12 in China mine collapse." *Anadolu Agency*. [Online]. Available: https://www.aa.com.tr/en/asia-pacific/death-toll-rises-to-12-in-china-mine-collapse/721344
6. "Photograph of the tunnel collapse extending toward the face," Fig. 16. *ResearchGate*. [Online]. Available: https://www.researchgate.net/figure/Photograph-of-the-tunnel-collapse-extending-toward-the-face_fig16_328392583
7. M. Allen, "Robots to the rescue: miniature robots offer new hope for search and rescue operations,"

Modern Diplomacy, Jul. 6, 2025. [Online]. Available: https://moderndiplomacy.eu/2025/07/06/robots-to-the-rescue-miniature-robots-offer-new-hope-for-search-and-rescue-operations/
8. Science Japan, "Beyond the Two Great Earthquakes: Building a Disaster-Resistant Nation with 'Tough' Robots 《Interview with Professor Satoshi Tadokoro of Tohoku University》," *Science Japan*, 2021. [Online]. Available: https://sj.jst.go.jp/stories/2021/s0615-01p.html
9. T. S. A. Attia, "Design and Development of a Novel Reconfigurable Wheeled Robot for Off-Road Applications," Master's thesis, Virginia Tech, Blacksburg, VA, 2018. [Online]. Available: https://www.researchgate.net/publication/337339078_Design_and_Development_of_a_Novel_Reconfigurable_Wheeled_Robot_for_Off-Road_Applications
10. The Japan Times, "Emergency teams to be equipped with unmanned water-cannon robots," *The Japan Times*, Dec. 16, 2024. [Online]. Available: https://www.japantimes.co.jp/news/2024/12/16/japan/japan-disaster-robots/
11. Arduino. "Arduino Documentation," *Arduino Docs*, 2025. [Online]. Available: https://docs.arduino.cc/
12. Raspberry Pi Ltd., "Raspberry Pi Documentation," *Raspberry Pi*, [Online]. Available: https://www.raspberrypi.com/documentation/
13. Raspberry Pi Ltd., "Buy a Raspberry Pi High Quality Camera," *Raspberry Pi* [Online]. Available: https://www.raspberrypi.com/products/raspberry-pi-high-quality-camera/
14. Melexis, *"MLX90640 32×24 IR array – Datasheet,"* 2019 (Rev. 12). [Online]. Available: https://www.melexis.com/en/documents/documentation/datasheets/datasheet-mlx90640
15. InvenSense (a TDK Corporation company), "ICM-42688-P 6-Axis MEMS MotionTracking® Device", Datasheet DS-000347 Rev. 1.8, Jul. 27 2023. [Online]. Available: https://invensense.tdk.com/products/motion-tracking/6-axis/icm-42688-p/
16. DFRobot, "A02YYUW Waterproof Ultrasonic Sensor (SKU SEN0311)," *DFRobot Wiki*, [Online]. Available: https://wiki.dfrobot.com/_A02YYUW_Waterproof_Ultrasonic_Sensor_SKU_SEN0311
17. Cytron 30 Amp 5 V–30 V DC Motor Driver (MD30C) — Cytron Technologies, "30Amp 5V-30V DC Motor Driver," [Online]. Available: https://www.cytron.io/p-30amp-5v-30v-dc-motor-driver